

Table identifying central key factors:

Importance (1-10)	Issue / Topic	positive aspects	negative aspects	Sources
	Anthropomorphism*	Motivated by: - Accessibility - Effective motivation - Social motivation	- The more human-like a system acts, the broader the expectations that people may have for it - ascribe implicitly mental capabilities to agents (that actually do not own), especially at the beginning, when agents responds correctly to user's needs and commands Uncanny valley: excessive similarity produces an abrupt descent of the sense of familiarity, until descend causing a sense of repulsion and disturbance in the user -> reduced realism inconsistency	(Guiot, 2019; Torresen, 2018)
7	Technology Acceptance Model	Factors that influencing adoption and trust: - Ease of Use - Perceived Usefulness		(Guiot, 2019)
	To increase trust	- Functional (complete required task) - Helpful (provide guidance, help) - Reliable (consistent and predictable) system		(Guiot, 2019, p. 120)
	Increased biases		- Gender: stereotypical attribution of personality traits - Culture: use female AI voice assistants	
10	Ethical challenges	- avoiding misuse and allowing for human inspection of the functionality of the algorithms and systems - when moving toward advanced autonomous systems, the systems should themselves be able to do ethical decision making to reduce the risk of unwanted behavior		(Torresen, 2018)

[illegible]

*attribution of human feelings, physical qualities and mental states to unanimated artefacts; humans tend to see their world as a reflection of themselves (Guiot, 2019, p. 115)

Further relevant articles:

- **Categorize social robot according to socially evocative, social interface, socially receptive, and sociable** -> design of the robot increases in its ability to support the social model in more complex environments
 - o Importance of social model, which is often applied to explain, understand, and predict behavior (Breazeal, 2003)
- Check presentation fidelity X interactivity with robot
 - o Scenario texts / mock-ups are not well suited to assess emotional / affective attitudes and social aspects (Xu et al., 2015)

Literature

Breazeal, C. (2003). Toward sociable robots. *Robotics and Autonomous Systems*, 42(3–4), 167–175.

[https://doi.org/10.1016/S0921-8890\(02\)00373-1](https://doi.org/10.1016/S0921-8890(02)00373-1)

Guiot, V. (2019). *Designing Trustworthy Agents*. Politecnico di Milano.

Torresen, J. (2018). A Review of Future and Ethical Perspectives of Robotics and AI. *Frontiers in Robotics and AI*, 4. <https://www.frontiersin.org/article/10.3389/frobt.2017.00075>

Xu, Q., Ng, J., Tan, O., Huang, Z., Tay, B., & Park, T. (2015). Methodological Issues in Scenario-Based Evaluation of Human–Robot Interaction. *International Journal of Social Robotics*, 7(2), 279–291.

<https://doi.org/10.1007/s12369-014-0248-9>